

The Impact of Climate Change on The Development of Mood Disorders

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Topic

Exploring the influence of climate change on global rates of clinical mood disorders.

Introduction

Rates of mood disorders among Americans have climbed for years as we simultaneously experience the increasingly negative effects of climate change. In particular, abrupt changes in temperature and weather conditions (as brought about by climate change) may serve to promote both the early onset and exacerbation of symptoms related to bipolar disorder. According to preliminary research, genetic inheritance accounts for nearly 80% of bipolar disorder cases (Barnett and Smoller, 2018), highlighting the condition as optimal for investigation via the fundamental ecological idea of gene-environment interaction.

Primary Objectives

1. Identifying the known interactions between climate and:
 - a. The expression of genes that may contribute to the development of mood disorders.
 - b. The fluctuation of bodily hormones that regulate mood.
 - c. The fluctuation of neurotransmitters that regulate mood.
2. Analyzing the influence and prevalence of genetic components (e.g., genes themselves, imbalances of hormones and neurotransmitters, etc.) that contribute to bipolar disorder, specifically.
3. Analyzing the prevalence of environmental factors that influence the prevalence of gene variants that contribute to bipolar disorder.

Literature Review

The goal of this research is to identify the potential relationships between changing environmental factors and the physiological components that lead to the risk of bipolar disorder. Due to the scope and intersectionality of this topic, the research done has been broken into three parts in order to draw conclusions between them. In order to explore the connection between the environment, bipolar disorder, and biological factors, the epigenetics of the biological factors that play a role in bipolar disorder can be explored alongside the base genetics of bipolar disorder and the environmental factors that can cause physiological change.

Researchers have completed numerous studies on the topic of gene-environment interactions. Considering mental health, the environment interacts with mood-related genes both directly and indirectly, typically via the mechanisms that underlie trauma and stress (Cianconi et al., 2020). Not only do natural disasters contribute to trauma and stress responses but extreme, prolonged heat does also. In the twenty-first century, as both natural disasters and extreme heat have become more frequent around the globe, the risk of environmentally derived stress and trauma increased as well (Diffenbaugh and Scherer, 2011). Of the more well-known psychiatric morbidities that accompany changing climates, seasonal affective disorder (SAD) diagnoses have increased significantly since the year 2000. In a study of U.S. college students in which researchers attempted to quantify the risk of psychiatric comorbidity among students with SAD, the researchers determined that 36.6% of students with SAD experienced continued psychiatric illness (Fonte and Coutinho, 2021). Therefore, among the studied college student population, more than one-third of students experienced longer-lasting psychiatric illness (e.g., general anxiety disorder, major depressive disorder, etc.) than was explainable by an SAD diagnosis alone. Considering SAD diagnoses as *direct* indicators of the impact of climate change on mental

health, the increasing frequency of extreme climate events was *indirectly* associated with a large increase in the incidence of general mental illness.

On the molecular scale, changing climate conditions directly affect the formation of bodily proteins as well as the balance of neurotransmitters within the brain. Extreme, prolonged heat—the most common and most stressful consequence of climate change—directly impacts the structural and functional integrity of a multitude of human proteins. When heat is present in excess of the discrete limit that a particular protein can tolerate, the protein denatures—unfolds, losing its function. Specialized proteins known as heat shock proteins (HSPs) are responsible for protecting human proteins from denaturation by heat. HSPs can refold destroyed proteins, allowing the human body to adapt to extreme heat. As the deleterious effects of climate change have increased (in recent history), the activity of HSPs has dwindled since heat also contributes to HSP malfunction and insufficiency. Considered the most important human heat shock protein, HSP 70 is involved in protecting neural proteins from destruction, and previous research has determined that its insufficiency is a key component in the development of neurodegenerative disorders (which include mental health disorders). Also of note, heat increases neural transmission speed. Prolonged exposure to extreme heat can lead to the deterioration of neurons and the development of disorders linked to neuronal hyperactivity, including bipolar disorder, epilepsy, and Alzheimer's disease (Habibi et al., 2014). While the extreme heat brought about by climate change may not directly affect gene structure and activation, heat directly affects the translated products of human genes—proteins. As the effects of climate change have worsened, HSPs have become more important in maintaining bodily homeostasis. However, heat also decreases the effectiveness of HSPs, highlighting the idea that the human body is unable to adapt to the rapidly worsening effects of climate change.

Extreme heat also affects several complex biological mechanisms in the human brain. Swedish researchers at the Centre for Occupational and Environmental Medicine found that extreme heat causes a stark increase in required energy by the heart to circulate blood since such heat destabilizes the thermoregulatory mechanisms of the human body. Many neurohormones and neurotransmitters are involved in these thermoregulatory mechanisms—those of note include serotonin, acetylcholine, dopamine, epinephrine, norepinephrine, and gamma-aminobutyric acid (GABA). Each of these molecules is fundamental in the development of mood, sleep, and sensation characteristics. When extreme heat causes thermoregulatory distress, these molecules fall out of homeostatic balance, contributing to malfunctions in typical mood, sleep, and sensation characteristics. Furthermore, mood, sleep, and sensation processes directly impact one another, so thermoregulatory imbalances cause multi-fold malfunctions in each process (Lohmus, 2018). Extreme heat, overall, is a major contributor to the increasing rates of psychiatric morbidity. Since some researchers have determined associations between climate change and the development of mental health conditions, the incidence rates of mental illnesses will continue to increase until global populations take significant actions to slow the progression of climate change.

In a 2006 paper by Panskepp, bipolar disorder, like most psychiatric disorders, is described to be a syndrome, or the interaction between multiple brain regions and neurochemical pathways with environmental stressors that cause a series of symptoms that can be classified as mood instability between manic and depressive episodes. The cause of bipolar disorder is broadly unknown, however, it is often associated with malfunction in the calcium signaling pathway as the release of calcium interacts with neural receptors and is sighted as playing a role in “nervous system excitability” (Harrison, Geddes, et al., 2018). Though bipolar disorder isn’t

extremely common, the global prevalence is estimated to be around 1-2% with variations being caused by the differing diagnosis rates in different countries (Rowland & Marwaha, 2018). It is also speculated that the rate of diagnosis varies greatly across socioeconomic class and different cultures as well. The inheritability of bipolar disorder is considered great at 9% which is about ten times more than the United States's national average (Barnett & Smoller, 2009). It is known that the genetic components of bipolar disorder are complex, as the disorder isn't inherited through classic Mendelian genetics, causing genome wide association studies (GWAS) to be the best way to study the transfer rate through generations. Since bipolar disorder manifests in mood changes, its inheritability can be confusing to say the least. Hormones are often cited as being a driving force behind emotions, and a 2023 study by Lyu, Zhao, et al., demonstrated that gonadal hormones (estradiol, progestogens, and testosterone) had significantly different levels in participants' brains when they were experiencing manic or depressive episodes. Similarly, another study was done where emotional behaviors were classified in mammals, and the different chemical levels in their brains were recorded for each emotional system (Panksepp, 2006). So, genes that regulate these chemicals and hormones could potentially be linked to the development of bipolar disorder. Recent GWAS have shown that the most common genes that are related to bipolar disorder are ones that relate to the functionality of neural ion channels, such as the ones activated by calcium and allow for the release of different chemicals in the brain (Barnett & Smoller, 2009). While the information about the details and potential causes of bipolar disorder is abundant, all of the literature cites the continued gaps and confusion around the disorder. It is clear that the disorder is inheritable, however it is unclear what exact genes are causing the symptoms, as well as what physiological changes truly cause the effects of the disorder. Most of the studies surrounding the topic are unable to provide a clear conclusion on these matters as

well, as they are simply association studies that are affected greatly by the underlying factors in the subjects as well as the vast unknown about neural conditions. Even though the literature in terms of bipolar disorder can leave a lot of speculation, it can provide a strong research-backed framework for the potential causes of the disease with room for further conclusions to be drawn from the synchronization of multiple sources.

While it is clear that more research is essential to understanding gene-environment interaction, there are certain studies that provide a solid foundation for the intersectionality between environment and genetic predisposition for bipolar disorder. One such experiment conducted by the Department of Psychiatry and Neuroscience at Dalhousie University aimed to find the correlation between prenatal and postnatal environmental stressors and genetic liability for severe mental illnesses including bipolar disorder. The study reported that polymorphism of the BDNF gene combined with environmental stress such as vitamin and nutrient deficiencies can result in the onset of bipolar disorder (Uher, 2014). While this is helpful, the study focused more on schizophrenia and acknowledged a need for further research on bipolar disorder. Environmental exposure may not necessarily be a ‘stressor’ leading to bipolar disorder, it may be something as common as sunlight. A study conducted at the University of New South Wales describes a seasonal pattern to episodes of mania and depression linked to changes in temperature, hours of sunlight, solar or ultraviolet radiation, barometric pressure, and more. While this study provides background explanation, it fails to provide meaningful association between seasonal patterns and hospitalizations due to mania as the rate of hospitalization does not provide an accurate estimation of manic episodes (Parker et al., 2017). Furthermore, a study was conducted to confirm the effects of solar insolation on the age of onset bipolar disorder. Patients with bipolar disorder from 36 collection sites spanning 23 countries at varying latitudes

were gathered. Each patient's family history was recorded as well as the age, polarity, and location of their first episode. The study concluded that increased sunlight in springtime and increased solar insolation greatly influenced the onset of bipolar disorder, thus confirming that location and changes in climate are an important factor to bipolar disorder. Limitations of this study include recall bias from patients (Bauer et al., 2023). An article published by Vanderbilt University explains how location in distance from the equator affects solar insolation, seasonal variations in sunlight, and disruption of circadian rhythms can contribute to increased or decreased occurrences of bipolar disorder within varying locations. This confirms that the climate of a location has a direct effect on the prevalence of inheriting and expressing genes for bipolar disorder (Rosenthal et al., 2020). For example, an experiment was conducted to evaluate seasonal disruptions of circadian rhythms and polymorphism of the PER3 gene, associated with bipolar disorder, across varying ethnicities. The study resulted in statistical differences between PER3 distributions in those with family history of bipolar disorder compared to those who do not although more research must be conducted on other variants of genes to further understand their connection to bipolar disorder (Aytac et al., 2022).

Research Implications

By establishing the multifaceted relationship between climate change and the development of mood disorders, the present research will contribute to widespread efforts to prevent the worsening of both global afflictions. More specifically, this research may influence corporations, governments, and political figures to better advocate for public policy aimed at reversing the progression of climate change. Similarly, our research serves as another explanatory model for the development of mood disorders and may improve efforts to curb the steadily increasing rates of mood disorders among global populations. Concerning the direct

effects of changing climate conditions, published research remains sparse. Because of the known correlations between changing climate conditions and mental health, more research into the genetic basis of these correlations is desperately needed.

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